

Assignment for Simulation and Modeling Lab — Day 1

Course Code: CSE 413

Level/Term: 4/1

Credit: 3

Instructor: Audity Ghosh

Submission Deadline: *(Instructor-defined)*

Submission Mode: Upload [.ipynb](#) file's github link to Google Classroom

Tasks to Complete:

1. Scalar & Array Operations (Redo Section with Your Own Values):

- Run the given code with scalar = [3.7](#) and a new array of at least 6 elements.
- Show outputs of all built-in mathematical functions (sin, cos, tan, etc.).

2. 2×2 Matrix Calculations (Already Done in Class):

- **Study and understand** the example code given for trigonometric, exponential, logarithmic, and rounding operations on a 2×2 matrix.

Task: Apply the Same Operations on a 3×2 Matrix

- Create a **3×2 matrix** with custom values.
- Perform and display results of:
 - Trigonometric functions
 - Exponential & logarithmic functions
 - Absolute value & square root
 - Remainder when divided by 2
 - Rounding (round, floor, ceil)

Make sure to clip inputs where needed to avoid math errors (e.g., arcsin/arccos domain issues).

4. Vector Functions

- Understand the operations on the 1D vector shown in the notebook.

Task: Perform Same Operations on a 2×3 Matrix

- Your 2×3 matrix can have **float values** (e.g., 2.1, 4.7...).
- Implement and display:
 - Max & Min (with indices)
 - Vector length
 - Sorted version (both **row-wise descending** and **entire matrix descending**)
 - Sum, Product
 - Median, Mean, Standard Deviation

Hint: Use `np.sort()` with `axis=1` for row-wise sorting. Use `.flatten()` for entire matrix sorting.

Bonus (Optional for Extra Marks):

- Try to **visualize** your matrix values using `seaborn.heatmap()` or `matplotlib`.
- Try replacing static inputs with **randomly generated numbers** using `np.random.uniform()` or `np.random.randint()`.

Checklist Before Submission:

- All code cells run without error.

- Outputs are clearly visible.
- You've added comments to explain key lines of code.
- The `.ipynb` file is named like this: `SIM_Lab1_YourName_StudentID.ipynb`